

Time and Consciousness: The Temporal Mechanics of the Conscious Mind

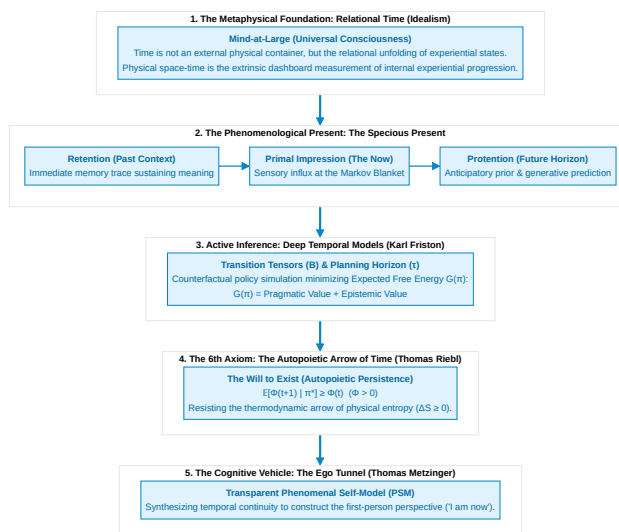
An Analytic Idealist and Active Inference Synthesis on the Specious Present, Deep Temporal Horizons, and the Autopoietic Arrow of Mind

Author: Thomas Riebl (Luxembourg)

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Theoretical Framework: The Conative-Integrative Framework (CIF)

Repository: <https://github.com/Thriebl/time-and-consciousness>



Executive Summary

The mystery of time has divided physics, neuroscience, and philosophy for centuries: 1. **The Physicalist Paradox:** Fundamental physics (from classical mechanics to general relativity and quantum mechanics) treats time symmetrically as a spatialized coordinate (t), largely indifferent to the unique status of the "Now" (the block universe view). 2. **The Phenomenological Reality:** Conscious experience is intrinsically directional, irreversible, and centered around a living present that possesses temporal width (*the Specious Present*). 3. **The Computational Requirement:** In Active Inference, agency and self-awareness are impossible without **temporal depth**—the capacity to simulate counterfactual trajectories through state-space.

This treatise demonstrates how the **Conative-Integrative Framework (CIF)** unifies these domains: * **Time in Analytic Idealism:** Physical time is the extrinsic appearance of relational transitions within *Mind-at-Large*. * **The Specious Present:** Formalized through Husserl's triad (*Retention* → *Primal Impression* → *Protention*) mapped directly into the predictive processing hierarchy. * **Temporal Depth as the Metric of Consciousness:** The transition from reactive automata (zero temporal depth) to conscious agents (deep temporal horizons optimizing $G(\pi)$). * **The 6th Axiom of Consciousness:** Establishing that conscious systems generate their own **autopoietic arrow of time**, actively preserving integrated causal power ($E[\Phi(t+1)] \geq \Phi(t)$) in defiance of thermodynamic decay.

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1. The Ontological Nature of Time in Analytic Idealism

In physicalism, time is often reified as an absolute external container or a geometric dimension of spacetime. In **Analytic Idealism** (Kastrup, 2019) and transcendental philosophy (Kant, 1781; Schopenhauer, 1819), time is recognized as the **intrinsic relational form of experiential succession**:

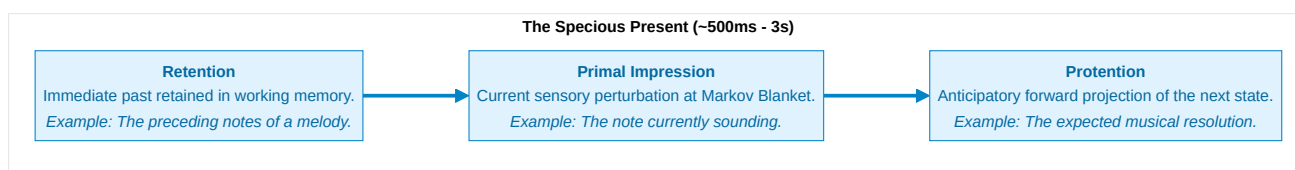
$$\text{Experiential State } \mathcal{E}_1 \xrightarrow{\text{Relational Transition}} \mathcal{E}_2 \xrightarrow{\text{Relational Transition}} \mathcal{E}_3$$

- **The Extrinsic Appearance:** What we measure with atomic clocks and observe as physical decay is the *extrinsic representation* (the dashboard dials) of the unfolding dynamics of Mind-at-Large across a dissociative boundary.
- **The Non-Dual Ground:** In its un-dissociated ground state, Mind-at-Large is atemporal (eternal presence / timeless potentiality). Time emerges as a localized operational property when a **Markov Blanket** dissociates an individual *alter* from the whole.

2. The Specious Present: Husserl, James, and Metzinger

Why do we not perceive reality as a sequence of disconnected, infinitesimal snapshots (t_0, t_1, t_2)?

William James (1890) coined the term “**Specious Present**” to denote that subjective time always has a duration (typically estimated between 500 ms and 3 seconds). Edmund Husserl (1928) formalized its tripartite phenomenological anatomy:



In the predictive brain (Metzinger, 2003, 2009; Friston, 2010): 1. **Retention corresponds to Bayesian priors and short-term synaptic memory traces** in cortical microcircuits. 2. **Primal Impression corresponds to the prediction error signal** generated at the sensory boundary. 3. **Protention corresponds to top-down generative predictions** projected forward in time.

The **Phenomenal Self-Model (PSM)** integrates these three temporal facets into a continuous, seamless “Ego Tunnel”—creating the felt sense of an abiding subject moving through time.

3. Deep Temporal Models: Active Inference and Counterfactual Agency

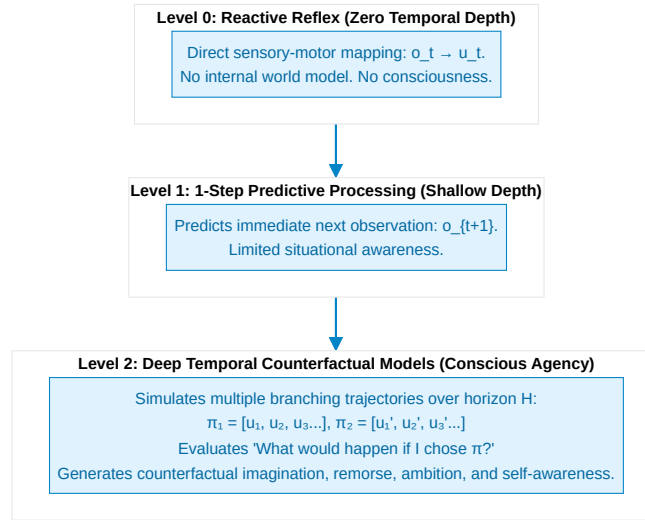
In the Free Energy Principle (FEP), living systems preserve their boundary by minimizing **Variational Free Energy** (F) and selecting actions that minimize **Expected Free Energy** (G):

$$\mathbf{G}(\pi) = \sum_{\tau=t+1}^{t+H} \mathbf{G}(\pi, \tau)$$

Where Expected Free Energy decomposes into **Pragmatic (Goal-Directed)** and **Epistemic (Information-Seeking)** value:

$$\mathbf{G}(\pi, \tau) = \underbrace{D_{\text{KL}}(Q(o_\tau | \pi) \parallel P(o_\tau))}_{\text{Pragmatic Value (Homeostatic Goal Pursuit)}} + \underbrace{\mathbb{E}_Q[\mathcal{H}[P(o_\tau | s_\tau)]]}_{\text{Epistemic Value (Exploration / Ambiguity Reduction)}}$$

The Temporal Depth Hierarchy:



Theorem (The Temporal Depth Condition for Consciousness):

A physical system cannot sustain phenomenal self-consciousness without generative transition tensors ($B = P(s_{t+1} | s_t, u)$) spanning a multi-step counterfactual planning horizon ($H > 1$). [1](#)

4. IIT 4.0: Temporal Grain and the Exclusion Postulate

In Integrated Information Theory 4.0 (Tononi et al., 2023), consciousness is quantified as **Integrated Cause-Effect Power (Φ)** across the Minimum Information Partition (MIP):

$$\Phi(M_1; M_2) = \frac{1}{2} \left(\ln \det(\Sigma_{M_1}) + \ln \det(\Sigma_{M_2}) - \ln \det(\Sigma_{\text{Whole}}) \right)$$

The Temporal Grain (τ):

IIT asserts that consciousness is not continuous at the Planck scale (10^{-43} s), nor at the geological scale, but exists strictly at the **spatial and temporal scale where integrated information reaches its absolute maximum**:

$$\tau^* = \arg \max_{\tau} \Phi(\tau) \approx 10\text{--}100 \text{ ms}$$

This temporal quantization explains: * **The Frame Rate of Consciousness**: Why human perceptual awareness resolves sensory streams in discrete micro-events (~20–50 Hz gamma/theta oscillations). * **The Exclusion Postulate**: Coarser or finer temporal grains have lower Φ and are excluded from subjective experience.

5. The 6th Axiom and the Thermodynamic Arrow of Time

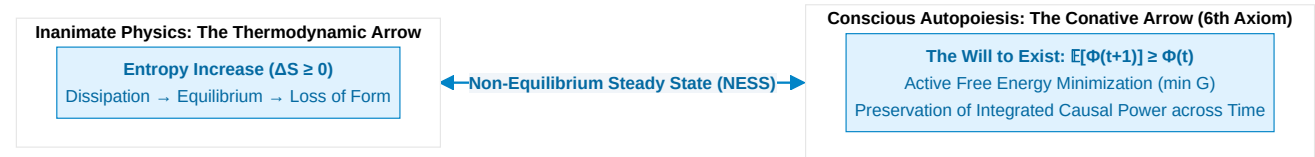
The fundamental law of inanimate physics is the **Second Law of Thermodynamics**:

$$\Delta S_{\text{universe}} \geq 0$$

Thermodynamic time is the irreversible increase in entropy, leading to dissipation, chaos, and thermal equilibrium (death).

In Thomas Riebl's **Conative-Integrative Framework (CIF)**, consciousness and biological autopoiesis are defined by the **6th Axiom ("The Will to Exist")**, which operates as a localized **anti-entropic arrow**:

$$\pi^* = \arg \min_{\pi} \sum_{\tau} G(\pi, \tau) \iff \mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t) \quad (\Phi > 0)$$



6. The Dissolution of Time: Minimal Phenomenal Experience (MPE) and Death

What happens to time when the cognitive mechanisms of the Ego Tunnel are dismantled?

1. Minimal Phenomenal Experience (MPE) / Mystical Awakening:

- In deep meditative absorption (Nirodha Samāpatti, non-dual awareness) or high-dose psychedelics, the top-down predictive projections (*Protention*) and narrative memory (*Retention*) collapse.
- The Markov Blanket becomes fully transparent.
- **Result:** The subjective illusion of flowing linear time collapses into the **Eternal Now (Nunc Stans)** of Mind-at-Large.

2. Biological Death (Physical Dissolution):

- The autopoietic action loop fails; $\mathbb{E}[\Phi(t+1)] \rightarrow 0$.
- The dissociative Markov Blanket dissolves.
- The localized temporal alter merges back into the unpartitioned, atemporal substrate of Mind-at-Large.

7. Conclusion and Open Research Horizons

Time and consciousness are not two independent entities meeting by chance. **Time as experienced is the operational signature of a dissociated conscious alter maintaining its existence against thermodynamic decay through deep active inference.**

Key Conclusions:

1. **Time is Idealistic & Relational:** Metric clock-time is an extrinsic dashboard reading of internal experiential dynamics.
2. **Consciousness Demands Temporal Depth:** Agency requires multi-step counterfactual generative models (B -tensors).
3. **The 6th Axiom defines Life & Mind:** The conative imperative ($\mathbb{E}[\Phi(t+1)] \geq \Phi(t)$) creates an anti-entropic temporal vector.

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1. **Scientific Context & Theoretical Attribution:** While Karl Friston et al. (2017, 2018) established temporal depth as a computational prerequisite for intentional action selection (*Planning as Inference*), and Anil Seth (2014, 2021) conceptualized “counterfactual richness” as a qualitative correlate of phenomenal presence, the *Conative-Integrative Framework (CIF)* by Thomas Riebl formalizes this insight for the first time as a strict mathematical **Theorem of Minimum Temporal Depth ($H > 1$) for Phenomenal Self-Consciousness**, directly coupled to the autopoietic preservation of Integrated Information (Φ) under the 6th Axiom ($\mathbb{E}[\Phi(t+1) \mid \pi^*] \geq \Phi(t)$). This theorem formally proves that purely reactive automata ($H = 0$) and myopic feedback loops ($H = 1$) suffer rapid phase-space and causal collapse ($\Phi \rightarrow 0$), establishing counterfactual temporal projection as a non-negotiable threshold of subjective mind. [↩](#)